Nathan Zhang

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Education

The University of British Columbia Sep 2023 – May 2028
B.A., Combined Major in Economics and Statistics, Minor in Data Science **Cumulative average: 93.5/100**
Selected coursework: Statistical Inference, Machine Learning and Data Mining, Advanced Econometrics, Time Series and Forecasting

Experience

ESG Research Analyst Co-op Sep 2026 – Present
British Columbia Investment Management Corporation (BCI) | Capital Markets & Credit Investments

- Investigate how executive compensation aligns management incentives with shareholder interests, helping the investment team prioritize governance risk factors. Independently construct variables from Capital IQ and ISS ECA and build Databricks pipelines for fundamental investment research.
- Automate voting rules, reconstruct ballot-time information, and reproduce research findings. Backtest against historical votes and use correlation analysis of subsequent risk outcomes to help the proxy team assess voting strategy.

Research Assistant Apr 2026 – Present
UBC Sauder School of Business | Supervisors: J. Frank Li and Cheng Chen

- As sole RA, build technological-exposure measures from CPS, O*NET, and USPTO data. Check that PatentsBERTa-based measures capture the intended meaning through Qwen comparisons and sampled manual reading.
- Examine how technological exposure relates to wages and occupational sorting, choosing fixed effects and fitting weighted FE OLS and GMM models. Draft the empirical section and maintain an R/Python replication package with Bash/Slurm HPC workflows so the analysis can be rerun.

Research Assistant May 2026 – Aug 2026
UBC Sauder School of Business | Supervisor: Allen Hu

- Checked whether a ProQuest corpus of 3.2M+ articles covered the financial news represented in RavenPack, using article-level matching to assess coverage and missingness.
- Built yt-dlp workflows that turn metadata into clean panels, organize and deduplicate video/audio, and check subtitles, preparing data for multimodal financial information extraction and analysis across media.

Research Assistant Jan 2025 – May 2026
Vancouver School of Economics at UBC | Supervisors: Jonathan Graves and Laura Nelson

- Led CV/NLP/NLI teaching proposals. Co-developed a Kouroi notebook testing whether image embeddings recover art historical categories, building ConvNeXt V2 classification workflows and writing method explanations.
- Co-developed a tutorial comparing zero-shot stance predictions with human labels, in first revision at Programming Historian. Designed and taught a computer vision workshop.

Selected Research

Sequential Tone Residualization and Abnormal Returns in Earnings Calls 2026
Accepted at ICAIF 2026, selected for oral presentation. Proposed an earnings-call study of whether management answers add return-relevant information beyond earnings surprise. Built the Capital IQ, IBES, and CRSP panel with Jerrold Huang and co-designed stage-specific information sets and dictionary/LLM validation. Time-holdout tests linked answer surprise positively to subsequent cumulative abnormal returns, with pre-event checks.

Memory Response Paths and Path-Structured Context Compilation 2026
Manuscript submitted, under review. Proposed response paths that compare matched futures with and without a memory to trace its effect on LLM answers. Built benchmarks and model API experiments and analyzed results across models and settings with Jerrold Huang. In a separate 144-item evaluation, PSCC improved LLM-judged quality by 6.9 percentage points over A-Mem with roughly one-third fewer API tokens.

Technical Skills

Programming & data: Python, R, SQL, Bash, Julia, PySpark, DuckDB, Polars, BQL
Methods: econometrics, factor models, causal inference, panel/time series, machine learning, NLP, computer vision
Libraries & platforms: NumPy, SciPy, scikit-learn, PyTorch, fixest, Databricks, Bloomberg, DevOps/Git, HPC/Slurm